

SHASHWAT MAHAJAN

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EDUCATION

Yeshwantrao Chavan College of Engineering, Nagpur

Expected June 2027

B.Tech in Computer Science Engineering | CGPA: **8.00 / 10**

Relevant Coursework: Machine Learning · Probability & Statistics · Linear Algebra · Data Structures & Algorithms · Database Systems · Computer Networks

Certification: Machine Learning Specialization — Codebasics.io (supervised & unsupervised learning, model evaluation, feature engineering)

EXPERIENCE

Generative AI Engineer Intern | Technical Hub — Generative AI Division

Jan 2026 – Present

- Designed and shipped LangChain-based RAG pipelines serving live end users; iterated on chunking strategy, embedding models, and re-ranking to drive a ~35% improvement in RAGAS answer relevance scores in production.
- Applied systematic prompt evaluation and failure analysis to reduce hallucination rates by ~25%; established retrieval-quality monitoring baselines (faithfulness, context precision) across multiple GenAI product initiatives.
- Collaborated with product teams to translate ML research findings into deployable pipeline changes, practising the full loop of hypothesis → experiment → evaluation → deployment.

PROJECTS

JobIntel Agent | Agentic AI Job Intelligence System

[GitHub](#)

Python · CrewAI · LangChain · Playwright · SendGrid · LinkedIn/Indeed/Naukri APIs · Vector Embeddings

- Architected a multi-agent CrewAI pipeline of four specialised agents — Scout, Analyst, Verifier, and Reporter — each with a defined role: sourcing jobs across LinkedIn, Indeed, Naukri, and career sites; scoring fit; validating recency and legitimacy; and ranking results by priority.
- Implemented intent-based job matching using semantic vector embeddings over role descriptions and JD text, surpassing keyword search by capturing contextual alignment between candidate profile and job requirements.
- Automated end-to-end delivery via SendGrid email report with ranked listings, relevance reasoning, and application links — compressing hours of manual search into a single scheduled pipeline run.

Financial Analyst RAG System | Production Research System

[GitHub](#)

Python · LangChain · ChromaDB · HuggingFace · RAGAS · AWS S3 · GitHub Actions

- Built a production RAG pipeline for real-time multi-company financial analysis; applied RAGAS to benchmark retrieval faithfulness and answer relevance — enabling data-driven iteration on chunking and retrieval strategy.
- Engineered HuggingFace query-expansion for cross-company comparison accuracy; implemented LLM output guardrails to constrain responses to grounded, cited content and reduce hallucination in high-stakes financial contexts.

Credit Risk Assessment System | Full-Stack ML

[GitHub](#)

Python · XGBoost · SMOTE · FastAPI · React.js · Docker · GitHub Actions

- Investigated class imbalance in a 10,000-record credit dataset; applied SMOTE oversampling and trained an XGBoost classifier (tuned via cross-validation) achieving 88% accuracy — a meaningful gain over the ~70% majority-class baseline.
- Architected a full inference pipeline (React.js → FastAPI → XGBoost) containerised with Docker and automated via GitHub Actions CI/CD, enabling reproducible zero-downtime deployments as a research-to-production artefact.

AWS Bedrock AgentCore Agent | Cloud-Native Agentic AI

[GitHub](#)

Python · LangGraph · AWS Bedrock · AWS Bedrock AgentCore · AgentCore Memory · FastAPI · Docker

- Built and deployed a stateful AI agent using AWS Bedrock AgentCore and LangGraph, orchestrating multi-step reasoning workflows with tool-calling and conditional branching across complex task sequences.

- Implemented AgentCore Memory for persistent cross-session context retrieval, enabling the agent to maintain user history and adapt responses based on prior interactions without re-prompting.
- Containerised agent runtimes with Docker and deployed cloud-native services on AWS infrastructure, achieving reproducible builds and environment parity between local development and production.

HACKATHONS

DevOnboard — AI Repository Intelligence Platform | IBM Hackathon | Individual

Python · LangChain · RAG · Vector Embeddings · GitHub API

- Individually built an AI platform that ingests any GitHub repo via URL, parses its structure and source code, and returns cited context-aware answers to developer onboarding questions — compressing ramp-up from hours to seconds.
- Engineered a chunking-and-embedding retrieval pipeline with semantic vector search for accurate Q&A across large multi-file codebases with full source attribution; presented live demo to IBM panel.

Smart Attendance System | Smart India Hackathon (SIH) | Team Lead, 6-Member Team

[GitHub](#)

MongoDB · Express.js · React.js · Node.js · Supabase · PostgreSQL · WebSockets

- Led architecture design, sprint planning, and task delegation for a 6-member team; architected a real-time WebSocket layer validated under 100+ concurrent users with zero data loss.
- Designed rolling QR-based authentication to prevent replay attacks; implemented MFA with Supabase Row-Level Security for strict per-user data isolation.

SITNOVATE — Real-Time Chat Engine | 24-Hour Hackathon | Individual

React · WebSocket · JWT

- Scoped and delivered a JWT-secured real-time chat system entirely within 24 hours; made rapid architecture trade-offs under time pressure and shipped a live demo to a judging panel.

TECHNICAL SKILLS

ML & AI: Supervised/Unsupervised Learning · XGBoost · Feature Engineering · Model Evaluation · RAG · LangChain · LangGraph · CrewAI · AWS Bedrock · AgentCore · Prompt Engineering · RAGAS · HuggingFace · ChromaDB · Vector Embeddings

Math Foundations: Probability · Statistics · Linear Algebra · Bayesian Reasoning · Loss Functions · Gradient Descent

Cloud & MLOps: AWS (Bedrock, AgentCore, S3) · Docker · GitHub Actions · CI/CD

Languages: Python (primary) · Java · JavaScript (ES6+) · SQL · C

Web & Systems: FastAPI · React.js · Node.js · PostgreSQL · MongoDB · REST APIs · WebSockets

ACHIEVEMENTS & CERTIFICATIONS

Competitive Programming: LeetCode Rating 1538 — 400+ problems solved (Python & Java) | CodeChef 2-Star

Certifications: Machine Learning Specialization — Codebasics.io | Full Stack Web Development (MERN Stack)